Mapping Chester

Vectors by Hand

This week you'll learn to split a vector into horizontal and vertical parts, add vectors, and find where you end up. You'll finish by drawing a route across Chester and finding how far you are from Subaru Park.

Plan your week

The whole packet takes about **2 hours**. That's about **25 minutes a day**. Write the day you'll do each part, and tick it off when it's done.

Part	What you'll do	About	Day I'll do it	Done
1	Warm-up: directions and signs (no calculator)	10 min		
2	Pythagorean theorem	15 min		
3	SOHCAHTOA: breaking a vector apart	20 min		
4	Adding vectors	15 min		
5	Finding direction	10 min		
6B	The Chester history route (map + table)	30 min		
6C	Your own route to school	15 min		
	Reflection + photos + submit	5 min		

> **Tips:** Do the parts in order; each one uses the last. Stuck for more than 10 minutes? Circle the problem, write one sentence about what you tried, move on, and message your teacher.

What you need

Page 2 has a ruler and protractor to cut out, in case you don't have your own.

Ruler with centimeters · protractor · drawing compass · pencil and 2 colored pencils · scientific calculator in **DEGREE** mode (check: sin(30) = 0.5) · a map of Chester on your phone or printed (Part 6C only)

Printing check. Print at 100% ("Actual size," not "Fit to page") so grid squares are 1 cm. This bar should measure exactly 10 cm:



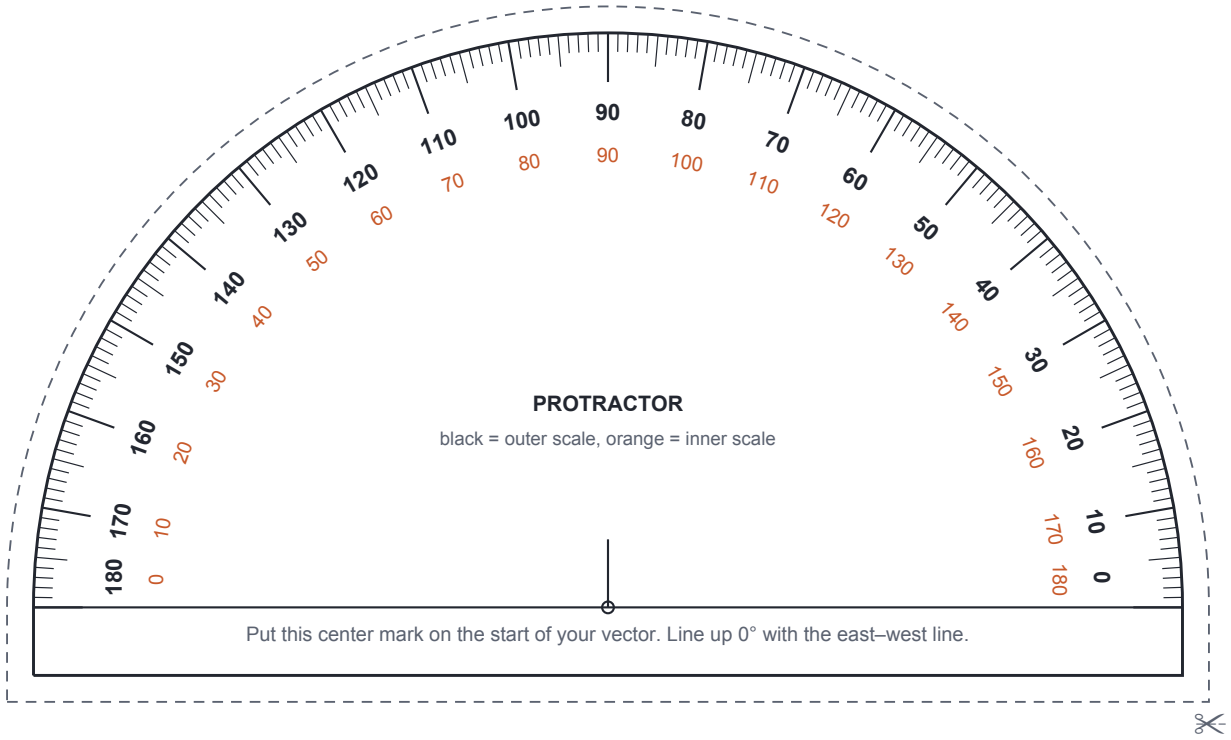
This bar should measure exactly 10 cm with your ruler.

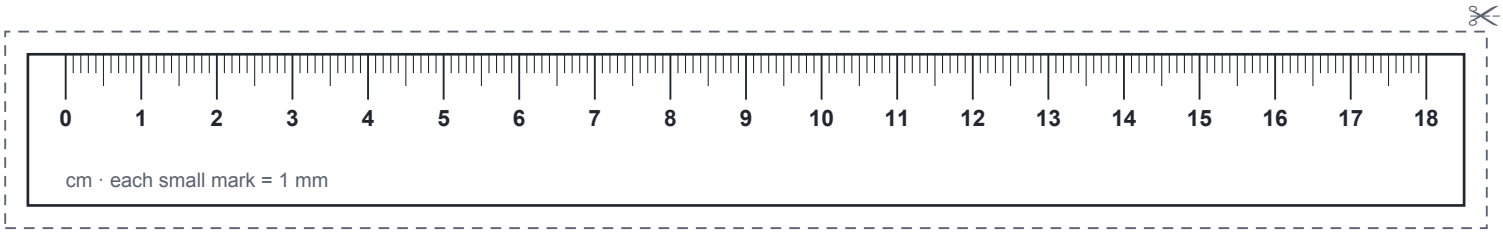
The rules

• Work by hand, in pencil.
• Every calculation shows the **formula**, the **numbers plugged in**, and the **answer with units**.
• When a problem says *measure*, write what you measured next to what you calculated. They should be close, not identical.

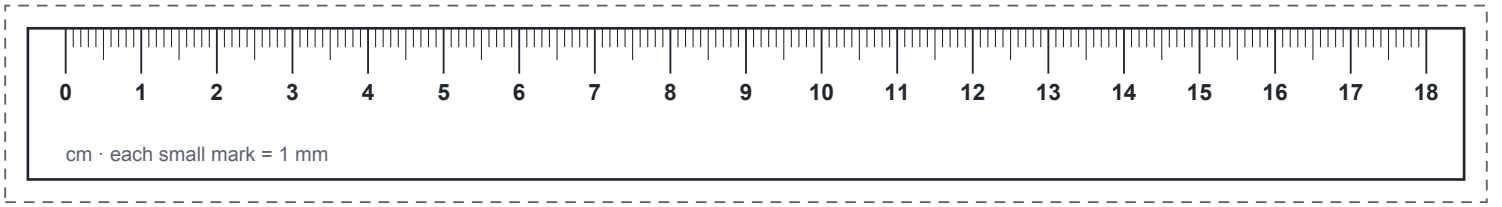
Cut-out tools: ruler and protractor

Cut along the dashed lines. Gluing this page to cardboard or a cereal box first makes the tools sturdier.
Check: the ruler's 0 to 10 should match the 10 cm bar on page 1. If it doesn't, reprint at 100% ("Actual size").





Spare ruler (or give one to a classmate):



Part 1 · Warm-up: Which way is that?

10 min · no calculator

1.1 In your own words, what is the difference between **distance** and **displacement**?

1.2 **"30° N of E"** means: face east, then turn 30° toward north. With your protractor, draw an arrow pointing 30° N of E. Then draw one pointing 30° E of N. Are they the same direction?

Same direction? YES / NO

1.3 In this activity, **east and north are positive (+)**, and west and south are negative (−). Write the sign for each: 4 km west ____ 2 km north ____ 7 m south ____

1.4 **Predict** (no math): you walk 3 blocks east, then 4 blocks north. Is your straight-line distance from the start **more than 7**, **exactly 7**, or **less than 7** blocks? Why?

Big idea: any diagonal vector can be replaced by a horizontal piece plus a vertical piece. Those pieces are called **components**. The rest of this packet is about finding them and using them.

Part 2 · Two perpendicular vectors 15 min

When two vectors are at 90° to each other, they make the two legs of a right triangle. The **resultant** R (the arrow from start to finish) is the hypotenuse.

$$R = \sqrt{(x^2 + y^2)}$$

Worked example. A Union winger runs 30 m north, then cuts 40 m east.
$R = \sqrt{(40^2 + 30^2)} = \sqrt{(1600 + 900)} = \sqrt{2500} =$ **50 m**
Drawing check: at 1 cm = 5 m, the legs are 8 cm and 6 cm, so R should measure 10 cm.

For each problem: **draw** it tip-to-tail on the graph paper (next page), **calculate** R here, and **measure** R with your ruler.

2.1 You walk 3 blocks east, then 4 blocks north. Scale: 1 cm = 1 block. Was your prediction in 1.4 right?

Work

Calculated R = _____ Measured R = _____

2.2 A bus leaves the Chester Transportation Center and drives 1.2 km east, then 0.5 km south. Scale: 1 cm = 0.1 km.

Work

Calculated R = _____ Measured R = _____

Drawings for Parts 2 and 3 (problems 2.1, 2.2, 3.2)

Name: _____ Write each problem's scale next to its drawing

Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").

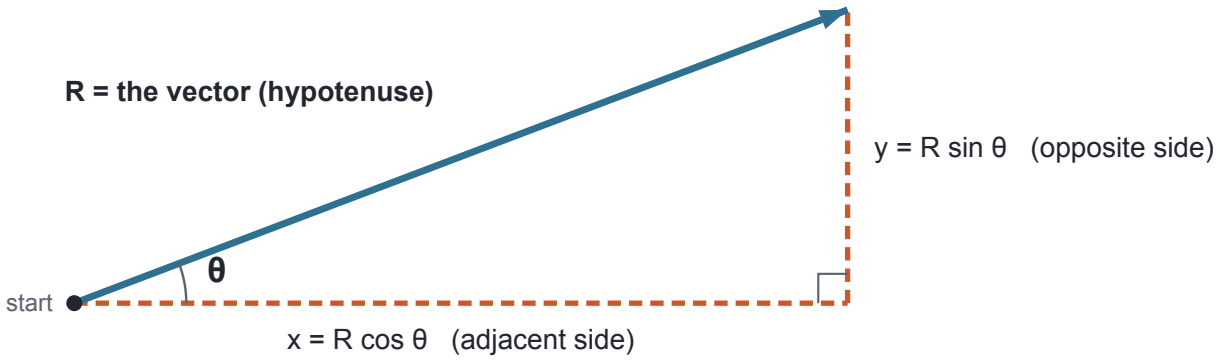
Part 3 · Breaking a vector apart: SOHCAHTOA

20 min

SOH	CAH	TOA
Sine = **O**pposite ÷ **H**ypotenuse	**C**osine = **A**djacent ÷ **H**ypotenuse	**T**angent = **O**pposite ÷ **A**djacent

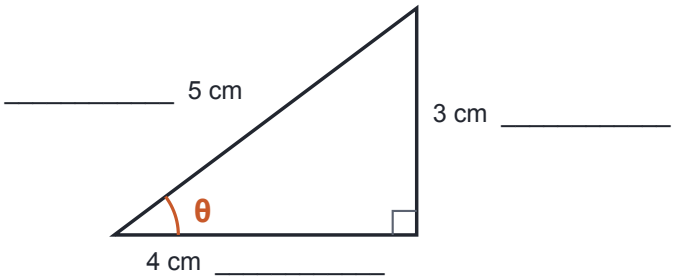
If you know a vector's size R and its angle θ from the east–west axis, its components are:

$$x = R\cos\theta \qquad y = R\sin\theta$$



Worked example. A goalkeeper punts the ball at 20 m/s, 30° above the ground.
Horizontal: x = 20 cos 30° = 20 × 0.866 = **17.3 m/s** Vertical: y = 20 sin 30° = 20 × 0.500 = **10.0 m/s**

3.1 On this triangle, write **opposite**, **adjacent** or **hypotenuse** on each blank (use the marked angle θ). Then write sin θ, cos θ and tan θ as decimals.



Printed to scale: measure it to check!

sin θ = _____ cos θ = _____ tan θ = _____

3.2 A student walks 10 m at 30° N of E. **(a)** Draw it on the graph paper (1 cm = 1 m). **(b)** Measure the x and y components with your ruler. **(c)** Calculate them with cos and sin.

	Measured	Calculated
x (east)		
y (north)		

Work

3.3 A free kick leaves the foot at 18 m/s, 35° above the ground. Sketch the triangle. Find the horizontal and vertical components.

Sketch + work

horizontal = _____ vertical = _____

Part 4 · Adding vectors

15 min

Step 1. Break every vector into x and y. **Step 2.** Add all the x's to get R_x, and all the y's to get R_y. Keep the signs! **Step 3.** $R = \sqrt{(R_x^2 + R_y^2)}$. Then check by drawing the vectors tip-to-tail.

Worked example. A = 3.0 km east. B = 2.0 km at 60° N of E.
A: x = 3.0, y = 0 B: x = 2.0 cos 60° = 1.00, y = 2.0 sin 60° = 1.73
R_x = 4.00 km R_y = 1.73 km $R = \sqrt{(4.00^2 + 1.73^2)}$ = **4.36 km**

4.1 A = 5.0 m at 30° N of E. B = 4.0 m north. Fill in the table and find R. Then draw it on the graph paper (1 cm = 1 m) and measure R.

Vector	x component (with sign)	y component (with sign)
A		
B		
Resultant	R_x =	R_y =

Work

Calculated R = _____ Measured R = _____

4.2 A walk around Chester: 2 km north, then 3 km east, then 1 km south. Find R by components, then by drawing (1 cm = 1 km).

Vector	x component (with sign)	y component (with sign)
2 km N		
3 km E		
1 km S		
Resultant	R_x =	R_y =

Work

Calculated R = _____ Measured R = _____

Part 5 · Which way? Finding direction 10 min

Use the **sizes** of the components (ignore the signs) to find the angle from the east–west axis: **$\theta = \tan^{-1}(|R_y| \div |R_x|)$**. Then use the signs to name it: **+x +y** = N of E · **−x +y** = N of W · **−x −y** = S of W · **+x −y** = S of E.

Example: R_x = 4.00, R_y = 1.73 → $\tan^{-1}(1.73 \div 4.00)$ = 23.4°, both + → **4.36 km at 23.4° N of E**.

For each row, calculate θ, then **measure it with your protractor** on your drawing.

#	From	R_x	R_y	θ calculated	Direction (__ of __)	θ measured
5.1	Problem 2.2					
5.2	Problem 4.1					
5.3	Problem 4.2					

Work

5.4 A vector has $R_x = -3$ m and $R_y = -3$ m. **Without a calculator**, give its direction. How did you know?

Drawings for Part 4 (problems 4.1, 4.2)

Name: _____ Write each problem's scale next to its drawing

Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").

Part 6 · Mapping Chester

45 min · split over 2 days

6B · The Chester history route (30 min)

A class trip starts at Subaru Park and makes three stops. Each leg is a straight-line ("as the crow flies") displacement. Stop positions are simplified for this activity, so they're close to the real places, but not exact.

Leg	From → To	Displacement
1	Subaru Park → the **Penn Landing site** on the waterfront, where William Penn first came ashore in Pennsylvania in 1682	1.80 km at 34° N of E
2	Penn Landing site → **Widener University**	2.25 km at 77° N of E
3	Widener University → the former **Crozer Theological Seminary** in Upland, where Dr. Martin Luther King Jr. studied from 1948 to 1951	1.97 km at 15° S of W

Use the **Chester route map sheet** in this packet. It already has the start, a north arrow and the scale (**1 cm = 0.25 km**). **Leg 1 is done for you** as a model. For legs 2 and 3:

1. **Convert** the length to cm: km ÷ 0.25 = cm.
2. **Draw** the leg with your protractor and ruler, starting at the tip of the last leg. Label the stop.
3. **Calculate** x and y with SOHCAHTOA. Leg 3 heads south and west, so both are negative.
4. **Add** them to your running R_x and R_y, then find your displacement from Subaru Park and its direction.
5. After leg 3, **draw a dashed arrow** (second color) from Subaru Park to the seminary. Measure it.

Leg	Given	Paper (cm)	x (km)	y (km)	Running R_x	Running R_y	Displacement from start	Total distance
1	1.80 km at 34° N of E	7.20	+1.49	+1.01	1.49	1.01	1.80 km at 34° N of E	1.80 km
2	2.25 km at 77° N of E							
3	1.97 km at 15° S of W							

Measured final arrow: _____ cm × 0.25 = _____ km, at _____° _____ of _____

Leg 1 (model). Paper: 1.80 ÷ 0.25 = 7.20 cm
x = 1.80 cos 34° = 1.80 × 0.829 = +1.49 km y = 1.80 sin 34° = 1.80 × 0.559 = +1.01 km
Displacement = √(1.49² + 1.01²) = 1.80 km $\tan^{-1}(1.01 \div 1.49) = 34°$, both + → **1.80 km at 34° N of E**

Leg 2

Paper length · x and y · running totals · displacement · direction

Leg 3 Remember: west and south are negative.

Paper length · x and y · running totals · displacement · direction

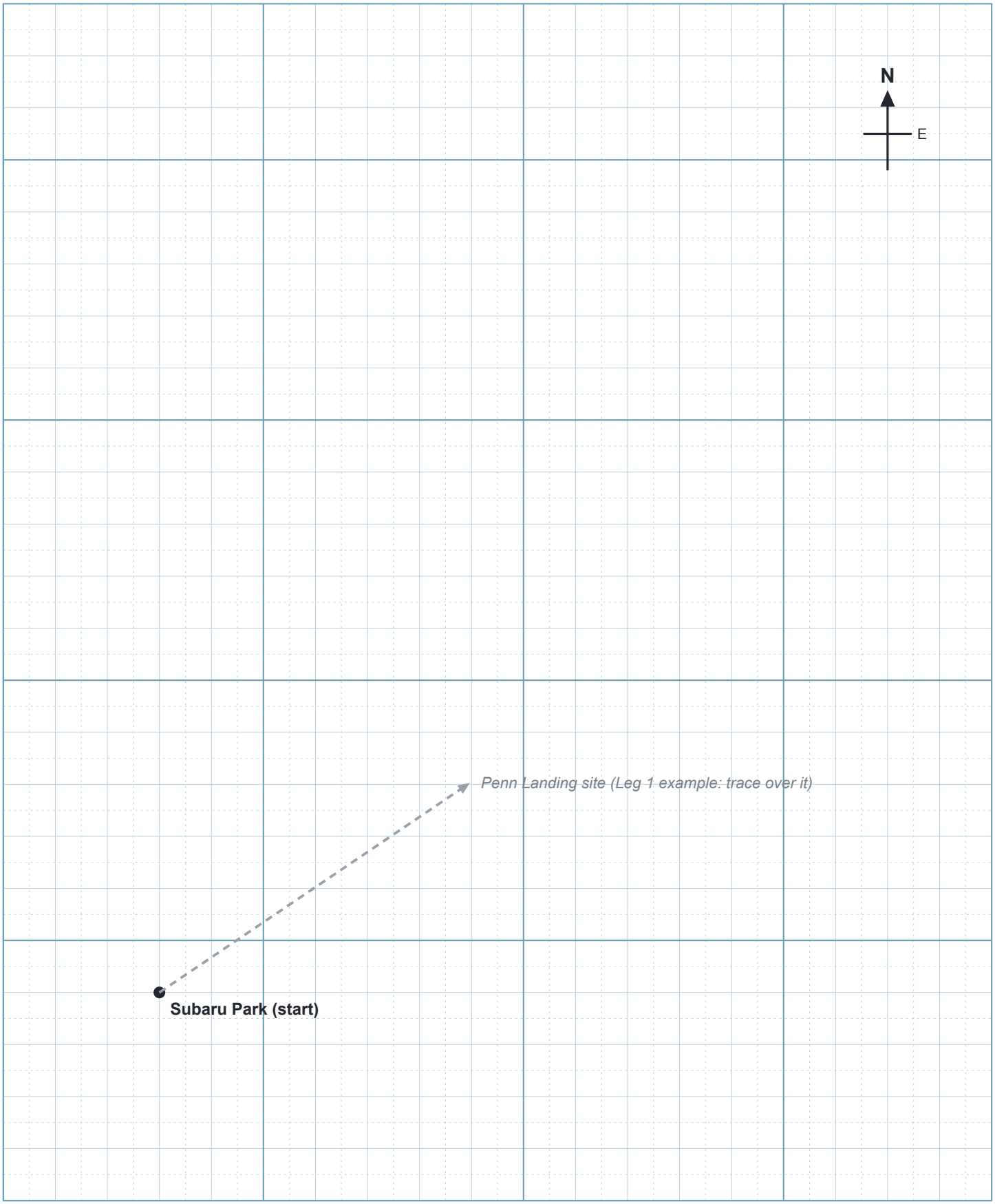
Q1 What is your **total distance** for the trip? What is your **displacement** from Subaru Park? Why is the displacement so much smaller? Point to your map in your answer.

Total distance = _____ Displacement = _____

Q2 The bus drives straight back to Subaru Park. What is the displacement for the whole round trip?

6B · Chester route map sheet

Name: _____ Scale: 1 cm = 0.25 km



Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size"). Solid arrows = legs. Dashed arrow (2nd color) = displacement from start.

6C · Your own route to school (15 min)

1. On a map app or printed map, pick a **start** you know (home, a store, a bus stop). Your **finish** is school.
2. Plan a route with **2 or 3 legs**. At least **1 leg must be diagonal** (not straight N–S or E–W).
3. Sketch the route on the graph paper on the next page. Copy the map's **scale bar** to find your scale.
   Example: a 500 m bar that measures 2.5 cm means 1 cm = 0.2 km.
4. Measure each leg's length and direction, fill in the table, and find your displacement from start to school.

Websites for planning your route (pick one):
• **Google Maps** (maps.google.com): on a computer, right-click your start, choose **Measure distance**, then click each corner of your route. Write down each leg's length.
• **OpenStreetMap** (openstreetmap.org): no account needed. Screenshot or print the map with the scale bar (bottom-left corner) showing.
• **Desmos** (desmos.com/calculator): optional check. Type your running totals as points, like (1.49, 1.01), to see your route on a grid.
The websites give lengths. You still measure each leg's **direction** with your protractor.

My scale: 1 cm = _____ km Start: _____

Leg	Length (km)	Direction	x (km)	y (km)	Running R_x	Running R_y	Displacement from start
1							
2							
3							

Show your work

Q3 Your map app shows a trip distance. Is it closer to your **total distance** or your **displacement**? Why?

Reflection and submit (5 min)

R1 Which part was hardest? What helped, or what are you still stuck on?

R2 Did you stick to your plan on page 1? What would you change next time?

Before you submit: name on every page · formula, numbers and units on every calculation · scale and north arrow on every drawing · clear photos of every page, uploaded **in order** to Canvas.

6C · My route map

Name: _____ My scale: 1 cm = _____ km

Sketch your route here. Copy the scale bar and mark your start and school.



Each small square = 0.5 cm. Each bold square = 5 cm. Print at 100% ("Actual size").